Teddy Liu

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EDUCATION

University of California, Davis | Davis, CA

B.S. Aerospace Engineering & B.S. Mechanical Engineering (Double Major) | Expected December 2027

EXPERIENCE

UC Davis Memorial Union Games Area | Davis, CA

Student Mechanic | November 2023 – Present

- Maintain, troubleshoot, and repair high-cycle industrial machinery, including electromechanical pinspotters and automated lane conditioning platforms.
- Diagnose mechanical drive assemblies, gearboxes, microswitches, and fluid delivery pumping systems using systematic fault-isolation techniques to minimize lane downtime.
- Inspect and service mechanical power transmission components, including belt drives, solenoid actuators, and bearings, ensuring continuous operational uptime across all lanes.
- Coordinate maintenance scheduling and machine status updates across the team, tracking repairs and flagging recurring electro-mechanical issues in real time.

Alameda County Public Works Agency | Hayward, CA

Engineering Intern | July 2022 - August 2022

- Utilized CAD software to catalog and map pedestrian ramp infrastructure across county road networks in support of ADA compliance surveying.
- Contributed to infrastructure documentation, providing accurate spatial data to support road improvement planning and civil engineering operations at the county level.

ACADEMIC & TECHNICAL PROJECTS

DIY Equatorial GoTo Telescope Mount Tracker

Mechanical Design / Mechatronics / Fabrication | April 2026 – Present

- Engineered an automated GoTo equatorial tracking mount for refractor and reflector telescopes, running OnStepX open-source firmware on an ESP32 microcontroller with TMC2209 silent stepper drivers.
- Designed and fabricated a GT2 belt reduction drivetrain for the Right Ascension (RA) axis, calculating the optimal reduction ratio to balance output torque with high-resolution sidereal tracking accuracy.
- 3D-modeled custom optical tube adapters in Autodesk Fusion 360, printed in Matte PLA with a high-perimeter gyroid infill pattern to maximize structural rigidity and eliminate flexure under dynamic payloads.

Automated Astrophotography & Instrumentation Pipeline

Optical Systems / Systems Integration / Signal Processing | May 2026 – Present

- Configured a headless Raspberry Pi 4B running KStars, Ekos, and AstroDMx Capture to orchestrate distributed camera control and image acquisition over a local wireless network.
- Integrated filter-modified DSLR and CMOS planetary cameras to maximize sensor dynamic range for low signal-to-noise deep-sky imaging.
- Developed a multi-stage image processing pipeline using Siril and RawTherapee for raw frame stacking, calibration, and radiometric correction, successfully characterizing and resolving a firmware compression artifact.

Automotive Electrical Systems & Security Integration

Automotive Electrical / Vehicle Systems | March 2026

- Integrated an aftermarket remote start and security system into a 2015 Toyota RAV4 ECU via direct wire splicing and circuit tracing, bypassing the need for prefabricated harness adapters.
- Hardwired a dual-channel dashcam into the vehicle's fuse box for continuous parking-mode surveillance, implementing low-voltage cutoff protection to maintain battery health.
- Performed vehicle-specific electrical modifications, including parking light circuit adaptation and manual door lock sensor integration; diagnosed and resolved a ground/power pigtail conflict.

Linux Systems Engineering & Homelab Infrastructure

Systems Engineering / Linux / Containerization | June 2025 – Present

- Deploy and administer a headless Linux server on x86 hardware, utilizing Docker to manage containerized workloads, localized web services, and dedicated game server environments.
- Manage reverse proxy routing, Wake-on-LAN remote access, and secure public tunnel exposure behind carrier-grade NAT, performing all system diagnostics and administration via the command line interface.

Optoisolated Network Wake Switch

Embedded Systems / Circuit Design / CAD | December 2025

- Developed a remote power management system using an ESP32 microcontroller to interface with a PC motherboard's front-panel power headers via PC817 optocouplers, ensuring galvanic isolation.
- Engineered a custom snap-fit enclosure in Fusion 360 utilizing precise dimensional tolerances to house microcontrollers, wiring terminals, and status LEDs, fabricated on an FDM 3D printer.

TECHNICAL SKILLS

- **Engineering Analysis & Design:** Autodesk Fusion 360, CAD Modeling, GD&T Fundamentals, Kinematic Analysis, DFM (Design for Manufacturing), Structural Optimization, Statics, Thermofluids, Fluid Mechanics.
- **Mechatronics & Embedded:** ESP32 Microcontrollers, Stepper Motor Control, TMC2209 Drivers, Actuators & Drivetrains, UART/I2C, Optoisolators, Automotive Electrical Integration, Soldering.
- **Fabrication & Prototyping:** FDM 3D Printing (Slicing, Infill & Perimeter Optimization), Wire Harness Assembly, Mechanical Assembly, Rapid Prototyping.

- **Software & Tools:** Python, C/C++, Bash, Linux (Ubuntu/Fedora CLI), Docker, Networking/Reverse Proxies, KStars/Ekos.